SPREADING THE FIBRE DIMENSION FROM THE GENERIC FIBRE TO A NONEMPTY OPEN OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8411-8417 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 8419-8432, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Closedness of the locus of bounded fibre dimension (Morphisms), and the two spreading lemmas for closed sets with empty or nonempty generic fibre (same chapter); cited inputs, with both shrinking steps written in the retained span.

Source reference. Original lemma 054W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6496–6501. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05F5: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6510–6515. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02FZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8432–8432. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume Y irreducible with generic point η and f of finite type. If X_η has dimension n, then there exists a nonempty open $V \subset Y$ such that for all $y \in V$ the fibre X_y has dimension n.*

Proof. Let $Z = \{x \in X \mid \dim_x(X_{f(x)}) > n\}$. By Morphisms, Lemma 02FZ this is a closed subset of X. By assumption $Z_\eta = \emptyset$. Hence by Lemma 054W we may shrink Y and assume that $Z = \emptyset$. Let $Z' = \{x \in X \mid \dim_x(X_{f(x)}) > n-1\} = \{x \in X \mid \dim_x(X_{f(x)}) = n\}$. As before this is a closed subset of X. By assumption we have $Z'_\eta \neq \emptyset$. Hence after shrinking Y we may assume that $Z' \to Y$ is surjective, see Lemma 05F5. Hence we win. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.